

PAI Lab Task: AI Aesthetic Stylist

Artificial Intelligence & Data Science | Semester 4

1. Project Overview

Is project mein humne ek **Aesthetic AI Stylist Chatbot** banaya hai jo Flask (Backend) aur HTML/CSS (Frontend) par mushtamil hai. Yeh bot users ko unki pasandida "vibe" ke mutabiq fashion suggestions aur styling ideas deta hai.

2. Technical Requirements

Technology	Usage
Python (Flask)	Server-side logic aur API handling ke liye.
HTML5 / CSS3	Structure aur professional aesthetic UI design ke liye.
JavaScript	Frontend aur Backend ke darmiyan real-time communication (Fetch API) ke liye.
Jinja2	HTML templates ko Flask ke saath render karne ke liye.

3. Folder Structure

Project ki files ki sahi placement niche di gayi hai:

```
PAI_Lab_Task/  
├── app.py (Main Python File)  
└── templates/ (Folder)
```

└─ index.html (Frontend File)

4. Backend Logic (app.py)

Backend mein humne ek knowledge_base banaya hai jo user ke keywords ko detect kar ke matching style details provide karta hai.

```
from flask import Flask, render_template, request, jsonify

app = Flask(__name__)

styles = {
    "minimalist": "Think neutral colors like beige, white, and black. Clean lines are key.",
    "vintage": "Go for 90s style denim and retro patterns. Very Pinterest-coded!",
    "streetwear": "Oversized hoodies, cargo pants, and chunky sneakers.",
    "aesthetic": "Focus on pastels and soft lighting vibes."
}

@app.route("/")
def index():
    return render_template("index.html")

@app.route("/chat", methods=["POST"])
def get_style():
    user_input = request.json.get("text").lower()
    response = styles.get("default", "Tell me a vibe like Minimalist or Vintage!")
    for key in styles:
        if key in user_input:
            response = styles[key]
            break
    return jsonify({"reply": response})
```

5. Troubleshooting (Common Errors)

- **TemplateNotFound Error:** Yeh error tab aata hai jab index.html file templates folder ke andar na ho.
- **File Rename:** File ka naam hamesha index.html hona chahiye, sirf index ya html nahi.

ISHMAL

- [redacted] or VS Code folder path na pehchane toh absolute path ka use karna behtar hai.

6. Conclusion

Yeh project AI aur Web Development ka ek basic magar effective combination hai. Ise portfolio aur LinkedIn par as a showcase project add kiya ja sakta hai.

